Capstone Project Proposal

One-Page Project Definition

Due: Week 8 | Points: 25

Instructions

Submit this one-pager for instructor approval by the end of Week 8. You will receive feedback before Phase 4 begins in Week 10. Both partners should contribute to this proposal—the problem definition should reflect both business and technical perspectives.

Business Partner: Angelyn Kornelsen **CS Partner:** Lesther Ouma

Project Title

Title: Revolutionizing Pricing Practices in a Small Christian Business

Problem Statement (2–3 sentences)

What problem are you solving? Who experiences this problem? Why does it matter?

Kornelsen Solutions Inc. needs a standardized job pricing system because pricing is currently done from memory and intuition with no documented formula. Currently, jobs are quoted inconsistently with variable prices for similar work and relationship-based adjustments that have no defined baseline which causes unpredictable revenue, wasted time and mental energy on every quote, and no visibility into exact profit margins per job. With data analysis, we can build a pricing formula that accounts for door type, size, materials, labor hours, travel distance, and customer tier which will reduce quoting time and effort to near zero, enforce price consistency across all jobs, and make profitability measurable on every single transaction.

Primary Stakeholder

Who will use your analysis/recommendations? (Be specific: a person, a role, an organization)

Royden Kornelsen, the founder and business owner of Kornelsen Solutions Inc.

Data Source(s)

What data will you use? Describe the dataset(s), source, approximate size, and how you will obtain access.

Note: Data must be real (not toy datasets). It may come from: campus operations, a local business, public datasets with genuine analytical questions, or your own data collection.

Data from Kornelsen Solutions Inc. QuickBooks Online invoices (around 500 invoices); Angelyn has access through QuickBooks online account.

Success Criteria (3–4 bullets)

How will you know the project succeeded? What specific outputs or insights should it produce?

1. If the program achieves baseline performance -> being able to quote the same way as business owner with 90% accuracy

2. If there's a 30% reduction in pricing inconsistency.

3. If we achieve a 90% task success rate with new users.

Scope Definition

Must-Have (in scope)	Should/Could Have	Won't-Have (out of scope)
Needs to have invoice information, needs to adjust output to the variables entered	The owner mentioned a time-tracking element, making it easier to estimate time to block out on future jobs.	Revenue or profit prediction

Key Risks

What could go wrong? What are you most worried about? (e.g., data access, data quality, scope creep, skill gaps)

- Data quality and relevance: Data from 10 years ago may not be able to accurately predict future pricing; data is scattered in different fields

Kingdom Business Connection
Why does this project matter beyond the grade? How does it connect to faithful stewardship, human flourishing, or serving others?

This capstone focuses on ensuring that customers are served justly and not overcharged. It reflects the biblical admonition in Leviticus to avoid dishonest standards in measuring length, weight, or quantity, and instead to use honest scales, weights, and measures. By upholding fairness and transparency, both the business owner and the customer can walk away without feeling exploited, preserving their sense of dignity and worth. At the same time, streamlining the quoting process reduces the mental load and time required for the business owner. This creates more space to invest in meaningful relationships, such as discipleship and other personal commitments.